4 The exterior angle of a regular n-sided polygon is 14° more than the exterior angle of a regular $(2n + 6)$-sided polygon.

(a) Write down and simplify an expression, in terms of n, each exterior angle of the $(2n + 6)$-sided polygon.

Answer ______________________ ° [1]

(b) Write down an equation in n and show that it reduces to $14n^2 - 138n - 1080 = 0$.

[3]

(c) Solve the equation $14n^2 - 138n - 1080 = 0$.

Answer ______________________________ [2]

(d) Explain why one of the solutions above is rejected.

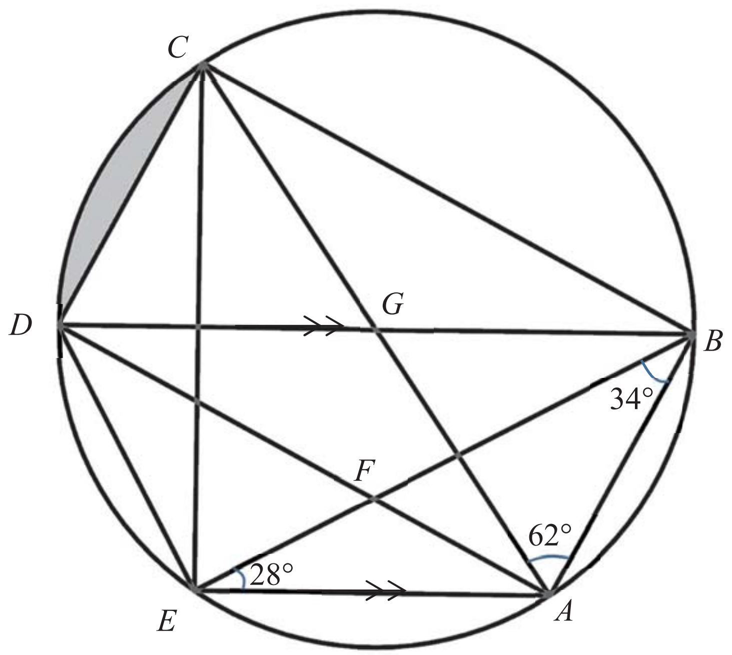
Answer

[1]

(e) Hence, find the value of each interior angle of the n-sided polygon.

Answer ________________________ ° [2]

5 The points *A, B, C, D* and *E* lie on the circle. $\angle AEB = 28^\circ$, $\angle CAB = 62^\circ$ and $\angle ABE = 34^\circ$.
AE is parallel to *BD*.



(a) Find two other angles that are 62°, stating the angle property.

Answer ______________________________ [2]

(b) Explain whether *AC* is the diameter of the circle.

Answer

[2]